

Proposal for a New Service Using AI: Accelerating Business with Data Analysis

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Current State of the Consumer Finance Industry

It has become an era where customers demand the 4 S's.

Speed

Review can be completed within a few minutes after application

Simplicity

Apply in just three simple steps on your smartphone.

Secret

Get a loan without your colleagues finding out

Safety

A trustworthy lending company

In order to improve profitability, the consumer finance industry

- wants to provide services to more customers while managing risk.
- wants to offer larger loans to customers with high creditworthiness.

Accurate **risk management** and identification of **low-risk customers** through customer data analysis are required

Utilize AI



From large volumes of information such as customer data, it is now possible to represent complex relationships that humans could not discover.

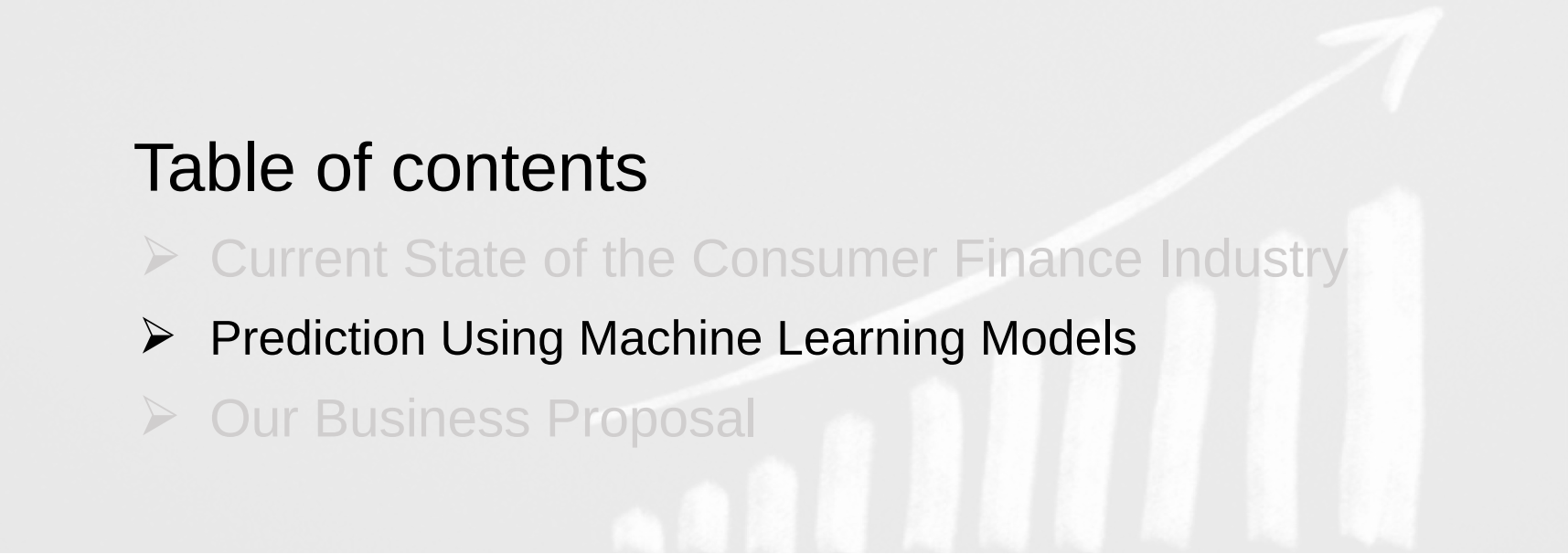
The background of the slide features a light gray bar chart with approximately ten bars of increasing height from left to right. A white arrow starts from the bottom left and points diagonally upwards towards the top right, passing over the bars.

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Introduction of Data

Approximately 200 types of customer information have been accumulated as

data

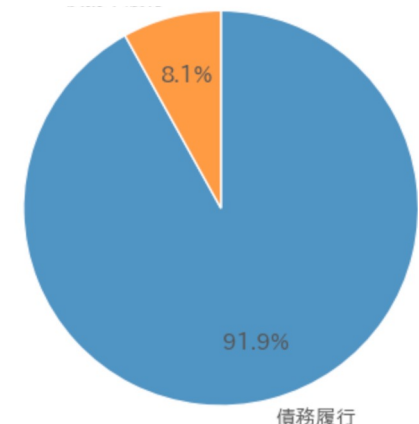
	SK_ID_CURR	TARGET	NAME_CONTRACT_TYPE	CODE_GENDER	FLAG_OWN_CAR	FLAG_OWN_REALTY	CNT_CHILDREN	AMT_INCOME_TOTAL	AMT_CREDIT	AMT_ANNUITY	...		
0	100002	1	Cash loans	M	N	Y	0	202500.0	406597.5	24700.5	...		
1	SK_ID_CURR	SK_ID_PREV	SK_ID_CURR	NAME_CONTRACT_TYPE	AMT_ANNUITY	AMT_APPLICATION	AMT_CREDIT	AMT_DOWN_PAYMENT	AMT_GOODS_PRICE	WEEKDAY_APPR_PROCESS_START			
2	0	2030495	271877	Consumer loans	1730.430	17145.0	17145.0	0.0	17145.0	SATURDAY			
3	1	2802425	108129	SK_ID_CURR	SK_ID_BUREAU	CREDIT_ACTIVE	CREDIT_CURRENCY	DAYS_CREDIT	CREDIT_DAY_OVERDUE	DAYS_CREDIT_ENDDATE	DAYS_ENDDATE_FACT	AMT_CREDIT_MAX_OVERDUE	
4	2	2523466	122040	0	215354	5714462	Closed	currency 1	-497	0	-153.0	-153.0	NaN
	3	2819243	176158	1	215354	5714463	Active	currency 1	-208	0	1075.0	NaN	NaN
	4	1784265	202054	2	215354	5714464	Active	currency 1	-203	0	528.0	NaN	NaN
				3	215354	5714465	Active	currency 1	-203	0	NaN	NaN	NaN
			4	215354	5714466	Active	currency 1	-629	0	1197.0	NaN	77674.5	

A portion of the data you provided

Data Contents

- Key data for each customer (gender, occupation, family information, presence/absence of default, etc.)
- Information on customers' past borrowings at other financial institutions (delinquent amounts, credit limits, etc.)
- Monthly balance status of customers
- Credit card usage information and withdrawal amounts for the month prior to the application date
- Key data at the time of past applications (contract type, application amount, purpose of use)
- POS cash balance information
- Installment payment information

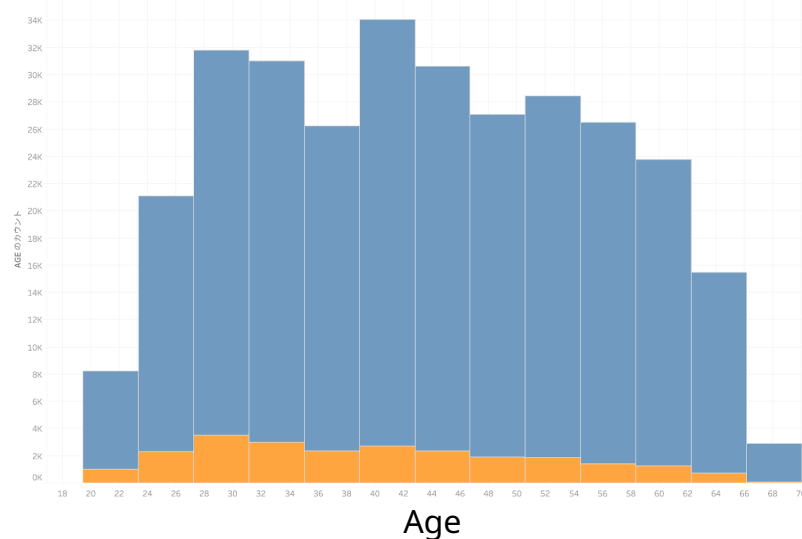
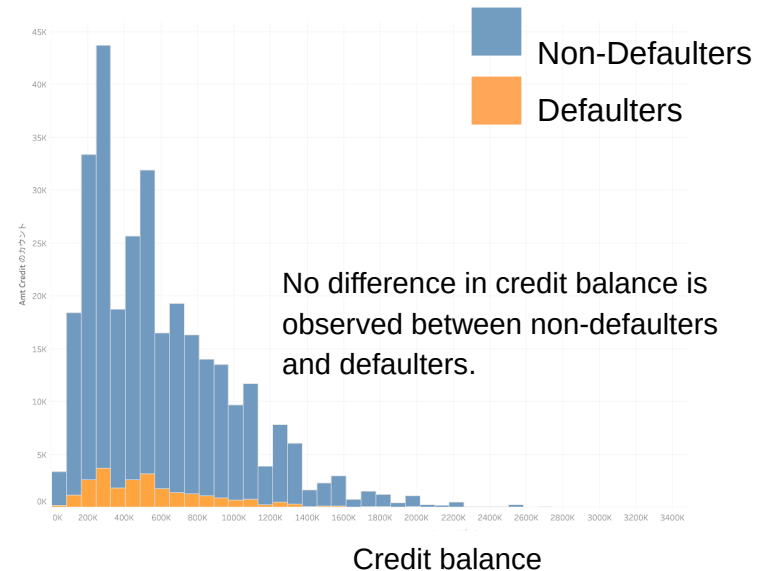
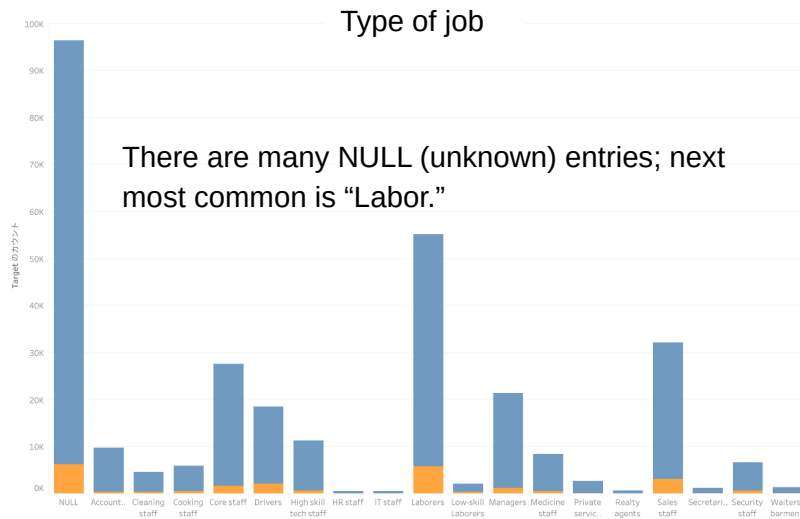
Proportion of Non-Defaulters



n= 307506

Data Visualization

We conducted various visualizations of the customer data.

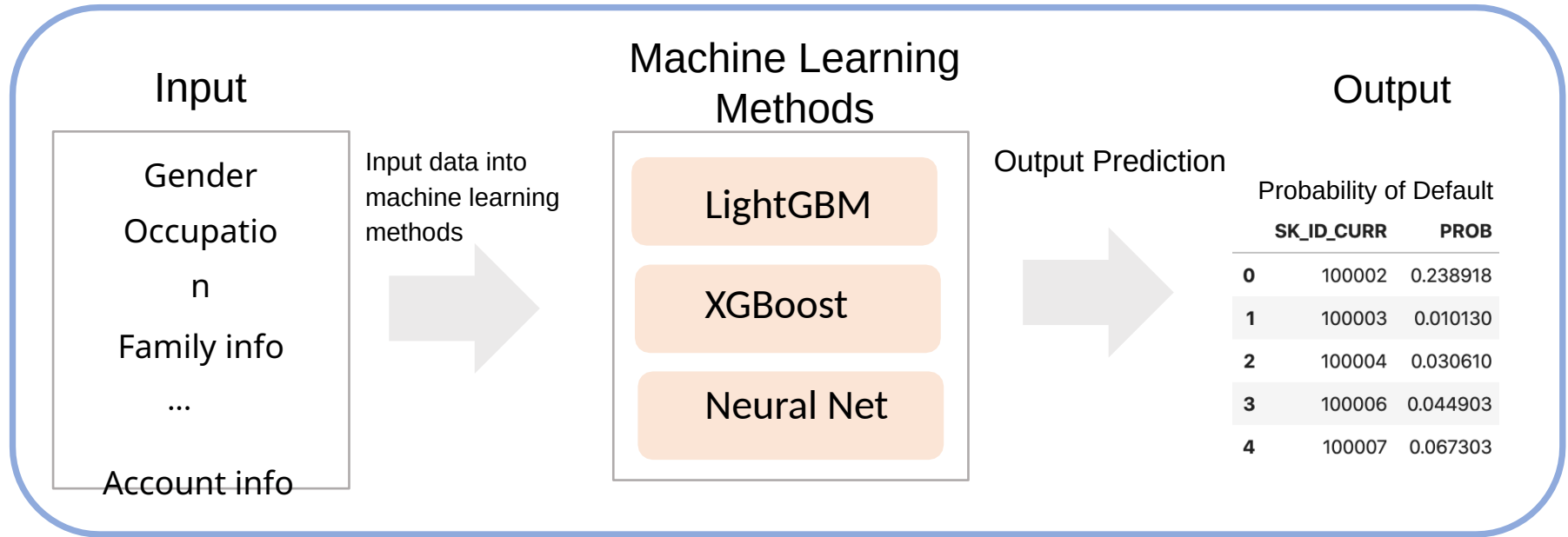


It is difficult to distinguish non-defaulters from defaulters using individual features alone.

Advanced analysis using machine learning methods is necessary.

Overview of the Machine Learning Model

From customer information such as gender, occupation, family information, and credit card usage amounts, the proposed model predicts the probability of default.



Defined as a credit score that is easy for humans to interpret*

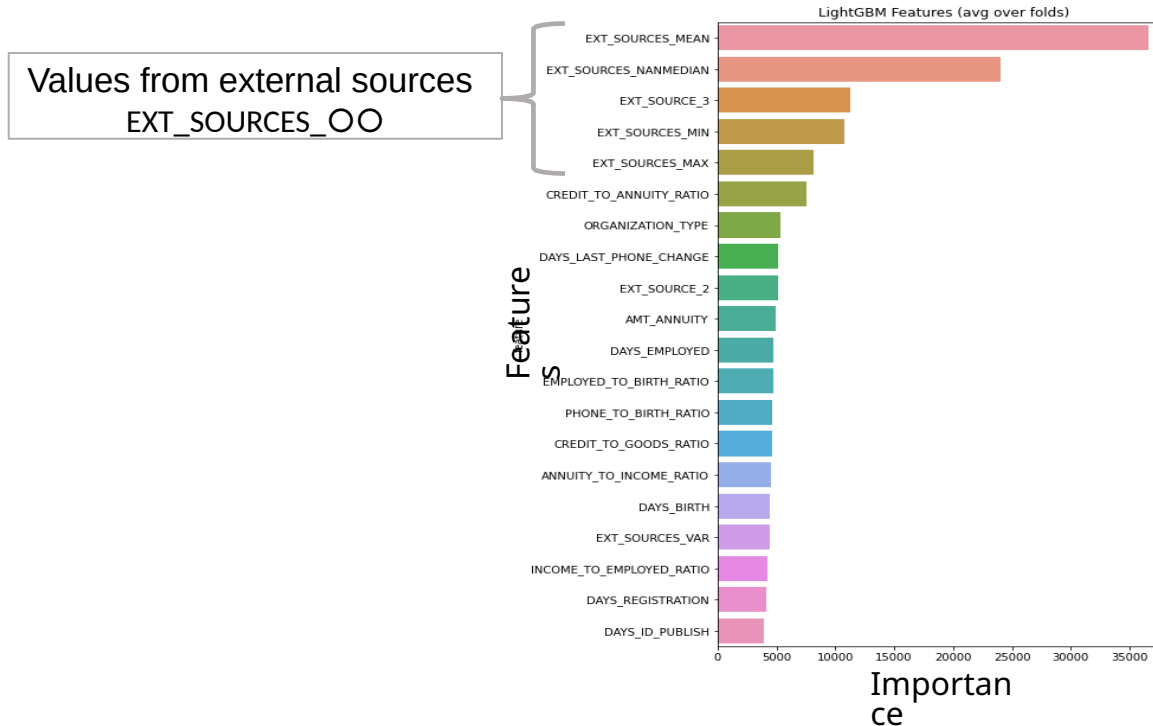
$$\text{Credit Score} = 1 - \sqrt{\text{Probability of default}}$$

SK_ID_CURR Credit Score		
0	100002	0.511208
1	100003	0.899353
2	100004	0.825043
3	100006	0.788097
4	100007	0.740571

*When calculating the credit score considering the distribution of default probability, a square root is taken.

Important Features for Prediction

Many of the features important for prediction are those related to values from external sources.



Features derived from processed data and statistical measures are important for AI predictions, indicating that the AI takes a different approach from human decision-making.

By utilizing both AI and human decision-making, highly accurate lending decisions become possible.

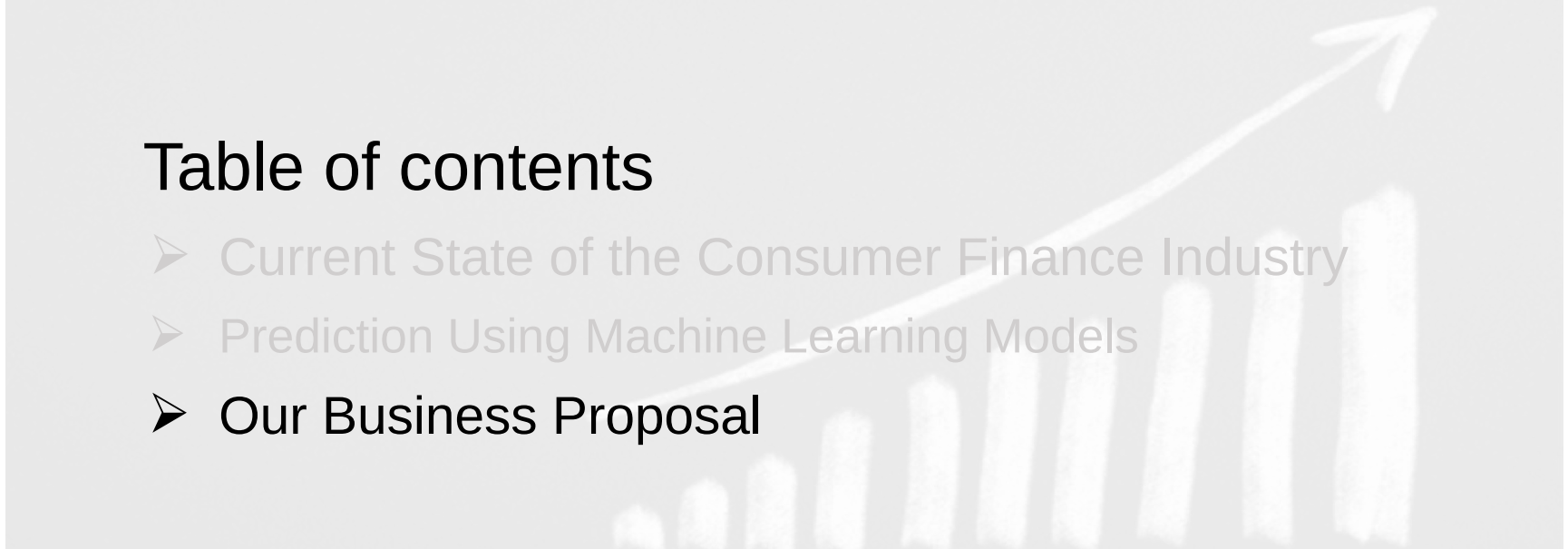
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Our Business Proposals

Business proposals utilizing credit scores and long-term support.

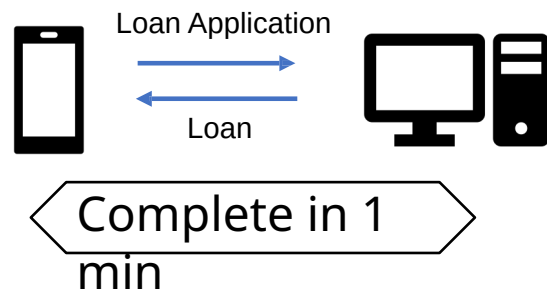
Businesses Utilizing Credit Scores

① Risk Management

Enables risk management for customers with a high likelihood of default

② Estimation of Low-Risk Customers

Enables proactive approaches to “low-risk customers” with repayment capability



Long-Term Support

Time Series Analysis

Enables lending decisions aligned with economic trends in each country

Model Updates

Through various analyses and training, further accuracy improvement is possible



① Risk Management Using Credits Scores

By utilizing AI, fast and accurate lending becomes possible.

Overview

Using the credit score, predict customers with a high probability of default.

Novelty

Decisions that were traditionally made based on human intuition and experience can be accelerated with higher accuracy.

Effects

Customer

Fast and responsive lending.

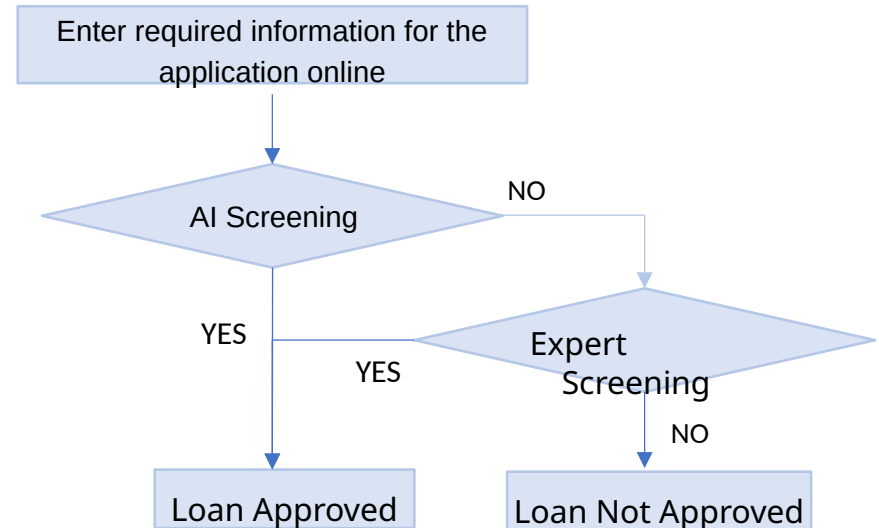
Company

Careful responses become possible for loans with a potential risk of default.

Point

Ideally, we would like to make “loan not approved” decisions based solely on AI screening; however, because there is a certain number of customers who can actually repay among those judged “NO” by AI, expert screening is necessary to lend to them.

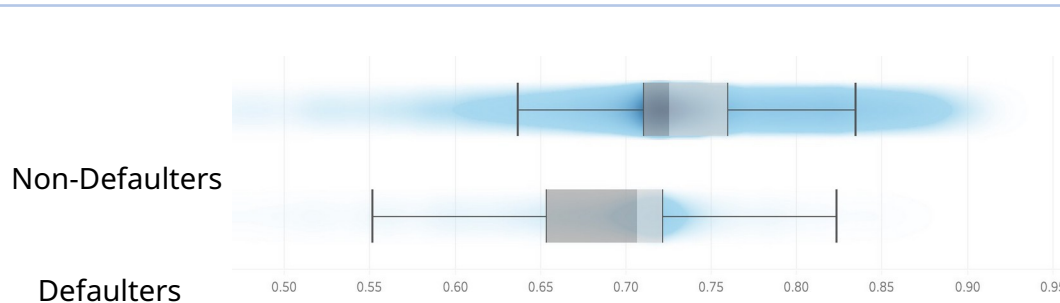
Loan Process Workflow



Validity of the Credit Score

Using the credit score, it is possible to distinguish non-defaulters and defaulters.

Difference in credit scores by non-default vs. default.



From the graph on the left, the credit score tends to be:
 Non-defaulters: Often 0.71 or higher
 Defaulters: Often below 0.71

If we define credit score $\geq 0.7 \rightarrow$ Non-defaulters

$< 0.7 \rightarrow$ Among defaulters,

Prediction
 Defaulter,

Number of Data		Prediction		Total
307506		○	×	
Results	○	237,881	44,801	282,682
	×	10,589	14,235	24,824
Total		248,470	59,036	

57%

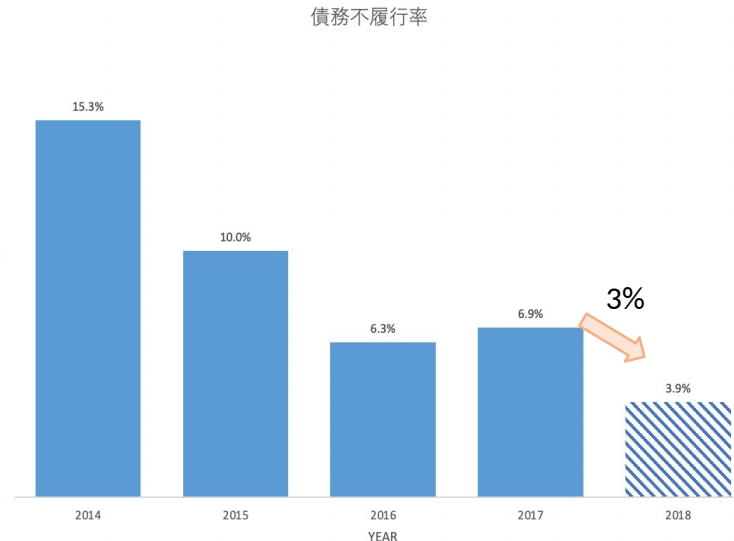
can be detected

On the other hand, 44,801 cases (14% of all data) were mistakenly predicted x;

Therefore, classification by experts with specialized knowledge is necessary.

Impact on Revenue

Reducing the customer's default rate leads to improved profitability.



Source :Home Credit annual report
2017

Assuming:

Median loan amount : 204,515 yen*

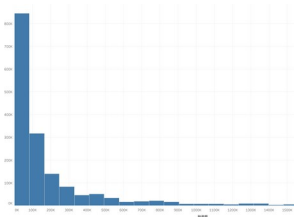
Annual number of loans: 41,608,720 people**

Difference in default rate: 3%,

annually,

Apx. 255.2 billion yen

can be reduced

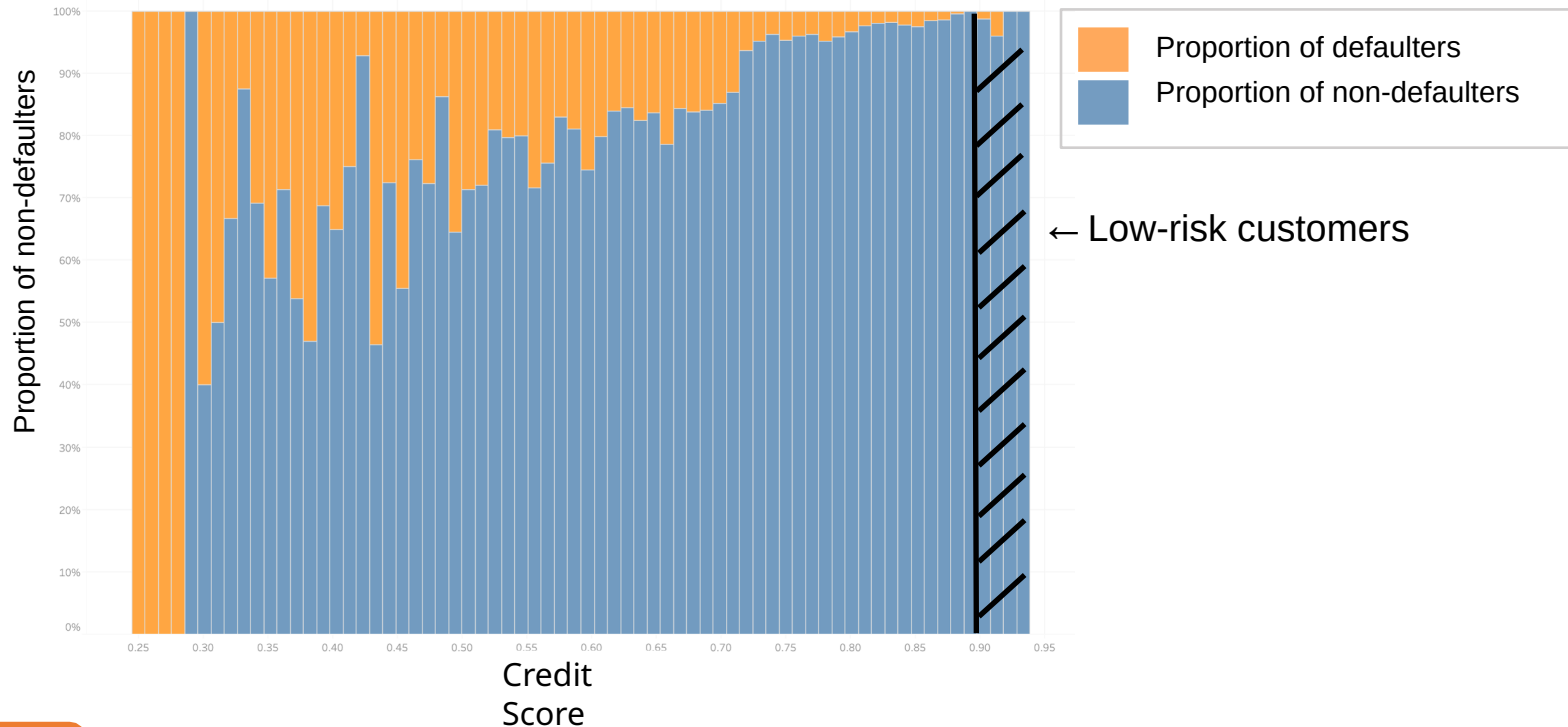


* As shown in the graph on the left, the distribution of loan amounts is right-skewed and the mean becomes large, so we estimate using the median (calculated with 1 tenge = 0.25 yen).

** Annual number of loans in 2017 listed on company's website was used

② Estimation of Low-Risk Customers by AI

It becomes possible to propose loan increases to customers whose probability of default is extremely low.



99.4%

of customers judged by AI to be low-risk*1 are non-defaulters.

Of the annual number of loans, 41,608,720 people*2, if we estimate low-risk customers at 5.5% of the total*3, then proactive approaches become possible to

Apx. 2.28million people

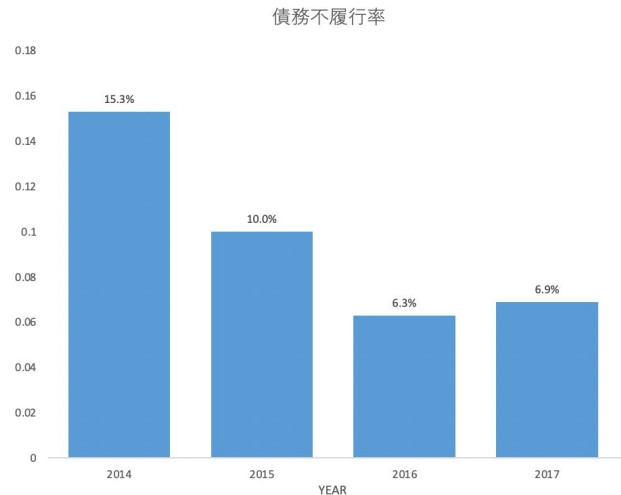
*1 Defined as customers with a credit score of 0.9 or higher

*2 Annual number of loans in 2017 listed on company's website was used

*3 Calculated from the data you provided
(17,035/307,506 = 0.055)

Long-Term Support

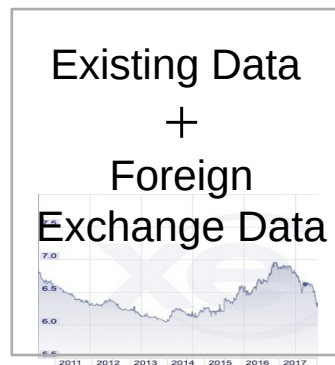
Through long-term consulting, lending that adapts to each country's financial markets and economic trends becomes possible.



Source: Partially excerpted from BTMU (China) Weekly Economic Report 2017

The 0.6% increase in the default rate in 2017 is attributed to the rapid growth of the consumer finance market due to yuan depreciation in China, while the credit system for China was insufficient.

→ To account for each country's economic trends, utilize foreign exchange data.



Input

Machine
Learning

Lending adapted to financial markets and economic trends becomes possible.

In addition, model updates can be implemented to improve the accuracy of lending decisions.

Quotation

In addition to the AI credit score, we propose long-term support.

AI Credit Score

Content

- Risk Management
Reduce the default rate by 57% and prevent an estimated 25.52 billion yen in annual losses.
- Estimation of Low-Risk Customers
Enable approaches to 2.28 million reliable customers with a non-default rate of 99.4%.

Price

We propose a fee of
30
million yen
for
providing AI.

This price corresponds to about 0.01% of the annual revenue improvement of 255.2 billion yen from risk management, plus the revenue from estimating low-risk customers.

To accelerate your business, our staff specializing in Finance × AI will utilize customer data to support you.

Long-Term Support

Content

- Time Series Analysis
Lending decisions can be made in line with economic trends.
- Model Updates
Introduce economic indicators and text mining to improve accuracy.

Price

We propose a consulting fee of
2.5 million
yen per month.